PROJECT PORTFOLIO

Proposed 3-Storied Residential Building

Architectural and structural drawing set

24 drawings | Architectural, structural, utilities and location

OWNER	Md. Milon Sarker, S/O Late Md. Ramiz Uddin
SITE	Mouza Kolabagan, Aishdhal, Kamarpukur, Saidpur, Nilphamari, Bangladesh
PLOT REFERENCE	Holding No. 1938, Khatiyan No. 2057, Dag No. 2491, J.L. No. 29
TOTAL LAND AREA	3.23 decimal
GROUND FLOOR AREA	883.55 sft
TOTAL FLOOR AREA	2,650.65 sft (as stated on the drawing sheet)
CONSULTANT	Landmark Engineering Solutions, Khoda Hafej Gate, Parbatipur Road, Saidpur, Nilphamari
PREPARED BY	Engr. Mobarok Hossain, B.Sc. Engg (WUB)
CHECKED BY	Engr. Saeed Sourav, B.Sc. Engg (Civil)

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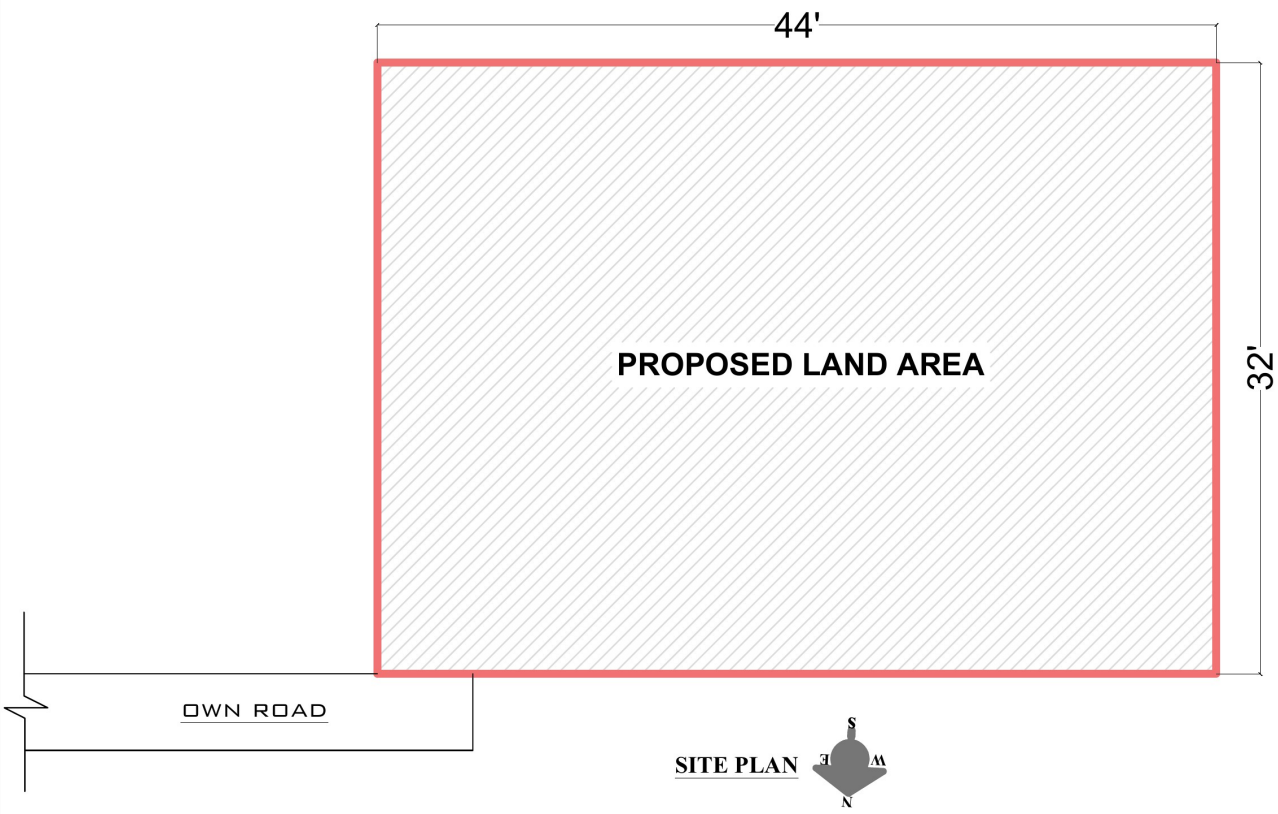
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Site Plan



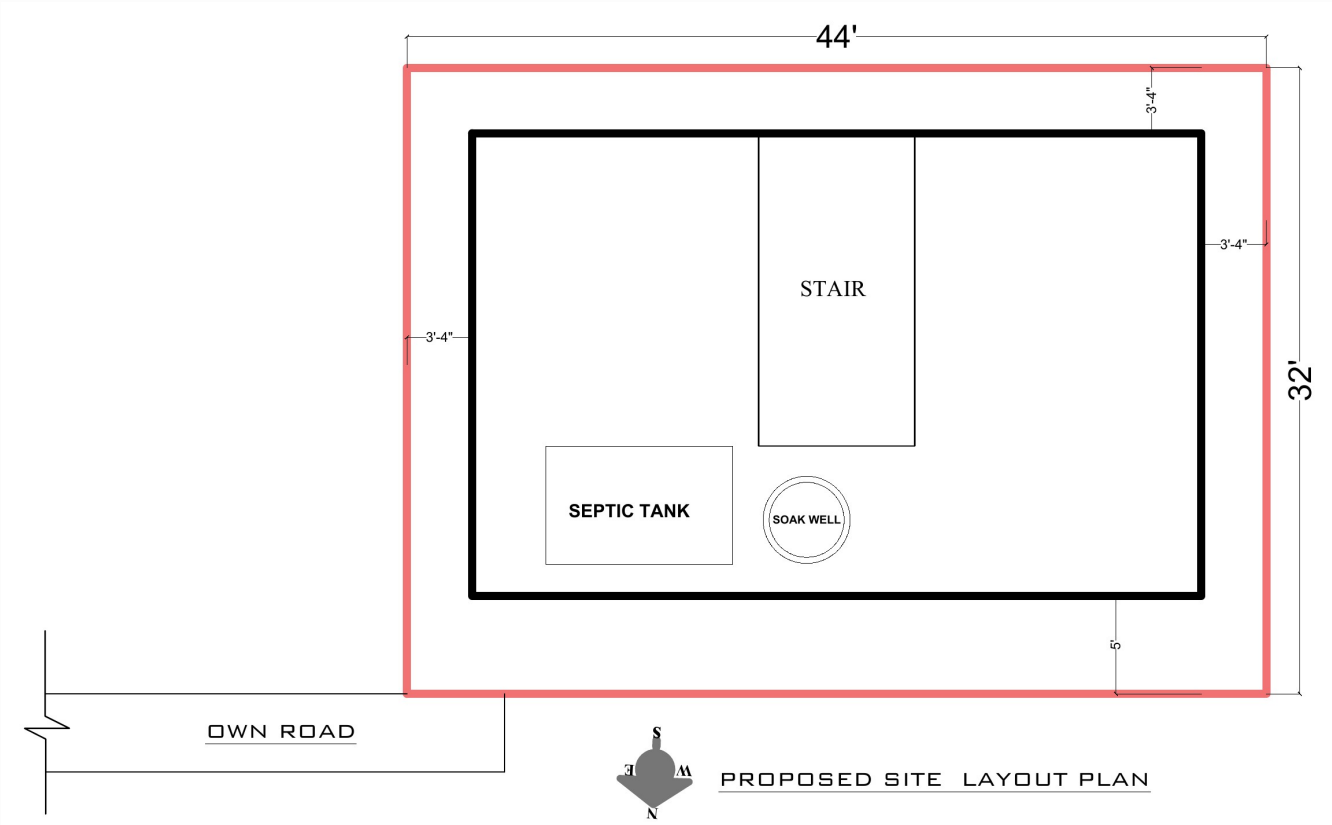
DESCRIPTION

Shows the proposed land area (hatched) as a rectangular plot beside the access road shown on the sheet as "Own Road". The plot outline is the basis for the building layout and setbacks.

KEY INFORMATION FROM THE DRAWING

- Plot size 44' x 32'
- Total land area 3.23 decimal (Holding 1938, Khatiyan 2057, Dag 2491, J.L. No. 29)
- Access from the own road along the plot edge

Proposed Site Layout Plan



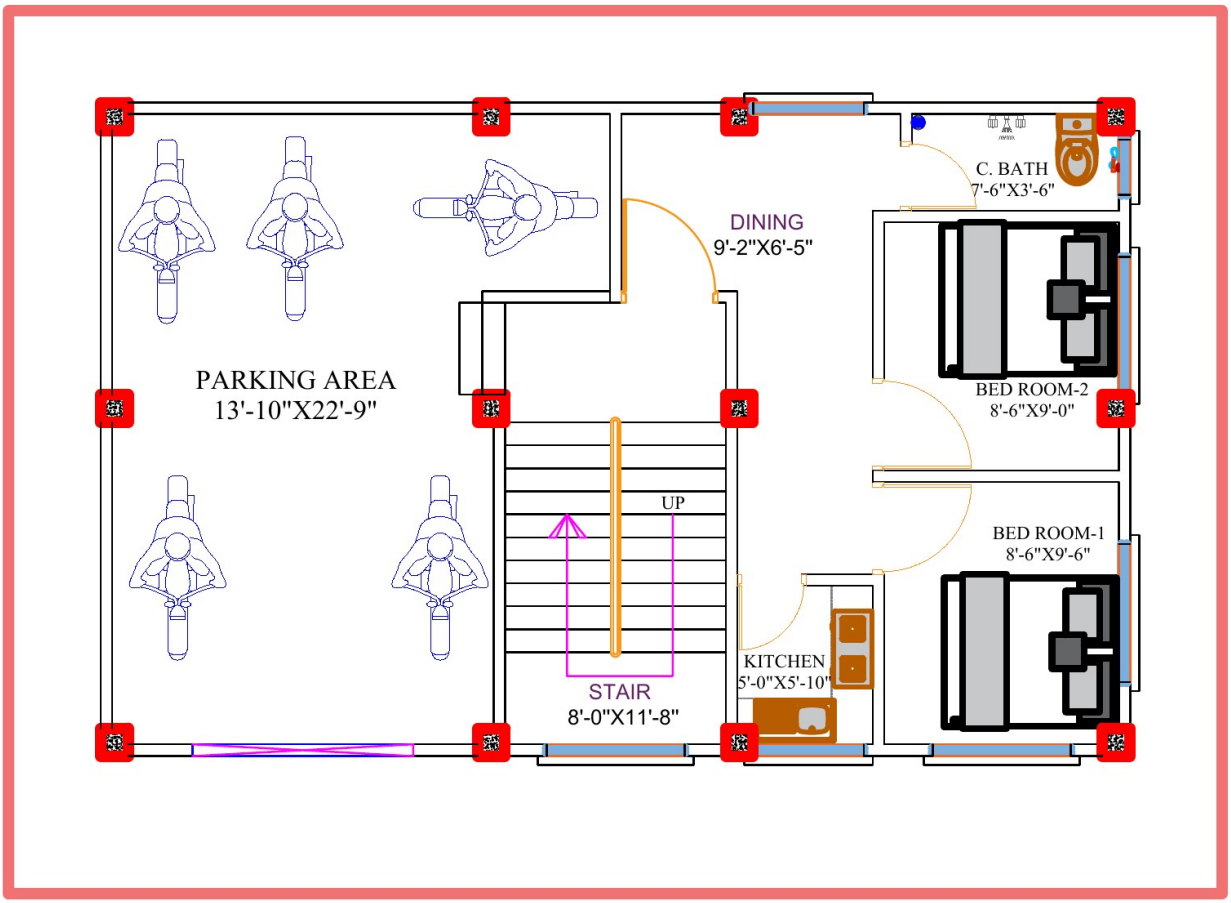
DESCRIPTION

Places the building footprint on the plot and shows the stair block and the septic tank and soak well in the open area beside the building.

KEY INFORMATION FROM THE DRAWING

- Plot 44' x 32'
- Setback dimensions of 3'-4" and 5' around the footprint
- Stair 8'-0" x 11'-8"
- Septic tank and soak well located within the plot

Ground Floor Plan



GROUND FLOOR PLAN

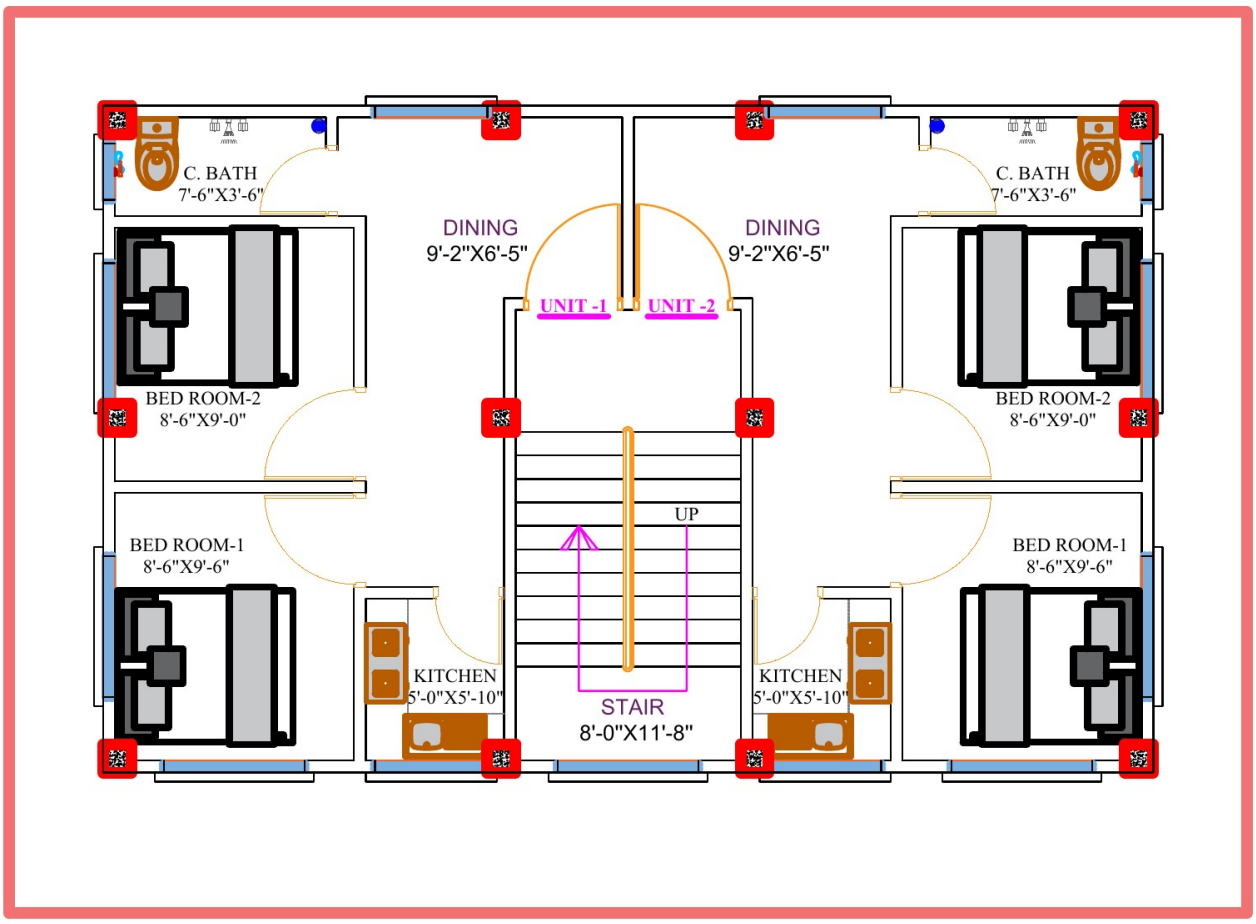
DESCRIPTION

Furnished ground floor plan showing the parking area, the stair and the residential rooms, with their sizes marked on the plan.

KEY INFORMATION FROM THE DRAWING

- Parking area 13'-10" x 22'-9"
- Stair 8'-0" x 11'-8"
- Bedroom-1 8'-6" x 9'-6" and bedroom-2 8'-6" x 9'-0"
- Kitchen 5'-0" x 5'-10", dining 9'-2" x 6'-5", common bath 7'-6" x 3'-6"

Typical Floor Plan



TYPICAL FLOOR PLAN

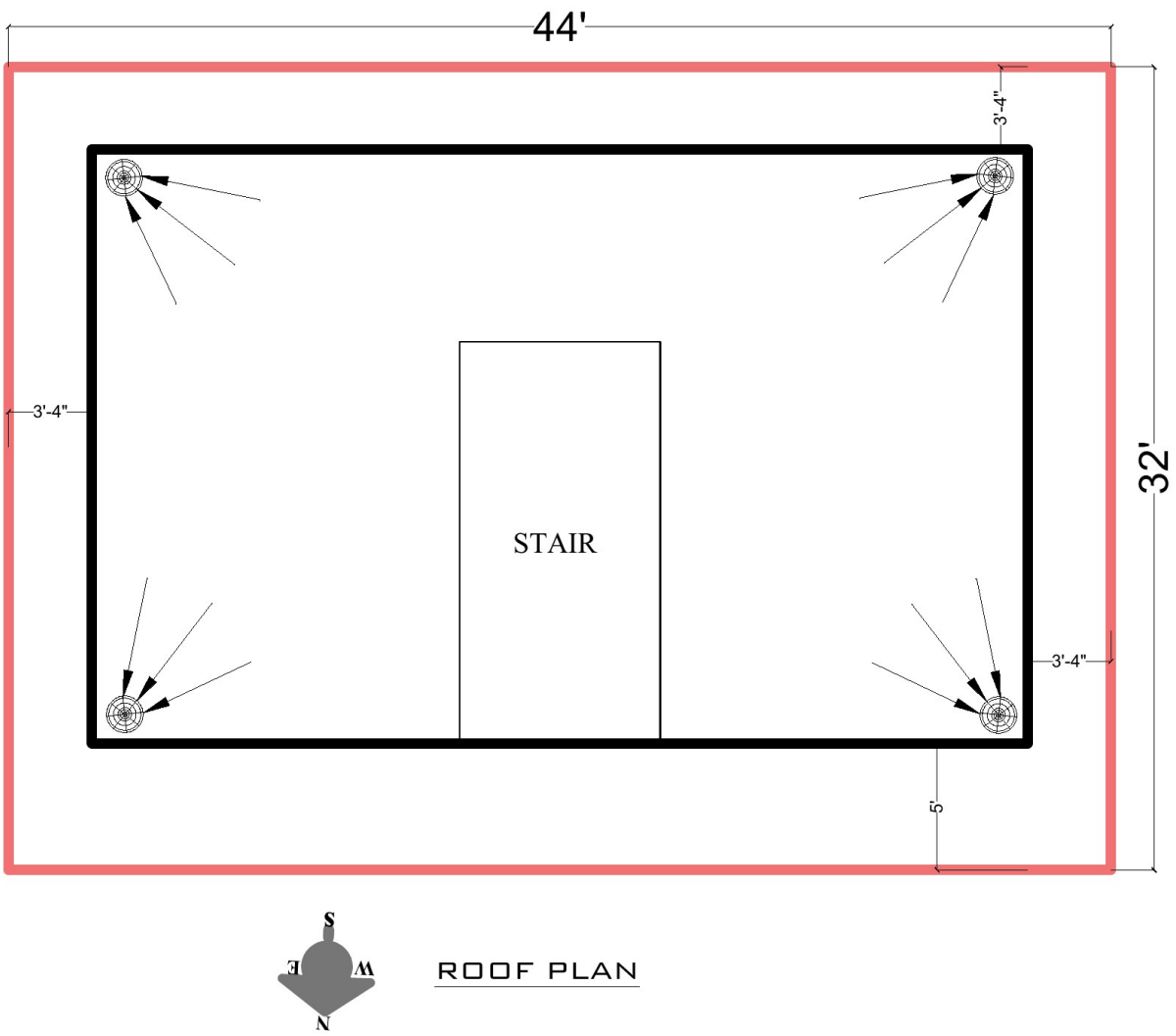
DESCRIPTION

Furnished plan of the repeating floor used for the 1st and 2nd floors. It holds two units (Unit-1 and Unit-2) on either side of the central stair, each with two bedrooms, kitchen, dining space and bath.

KEY INFORMATION FROM THE DRAWING

- Bedroom-1 8'-6" x 9'-6" and bedroom-2 8'-6" x 9'-0" in each unit
- Kitchen 5'-0" x 5'-10", dining 9'-2" x 6'-5"
- Common bath 7'-6" x 3'-6" in each unit
- Central stair 8'-0" x 11'-8"

Roof Plan



DESCRIPTION

Plan view of the roof slab with the stair head-room outline, drawn on the same plot outline as the layout plan.

KEY INFORMATION FROM THE DRAWING

- Plot 44' x 32'
- Stair head-room 8'-0" x 11'-8" marked
- Setback dimensions of 3'-4" and 5'

Front Side Elevation



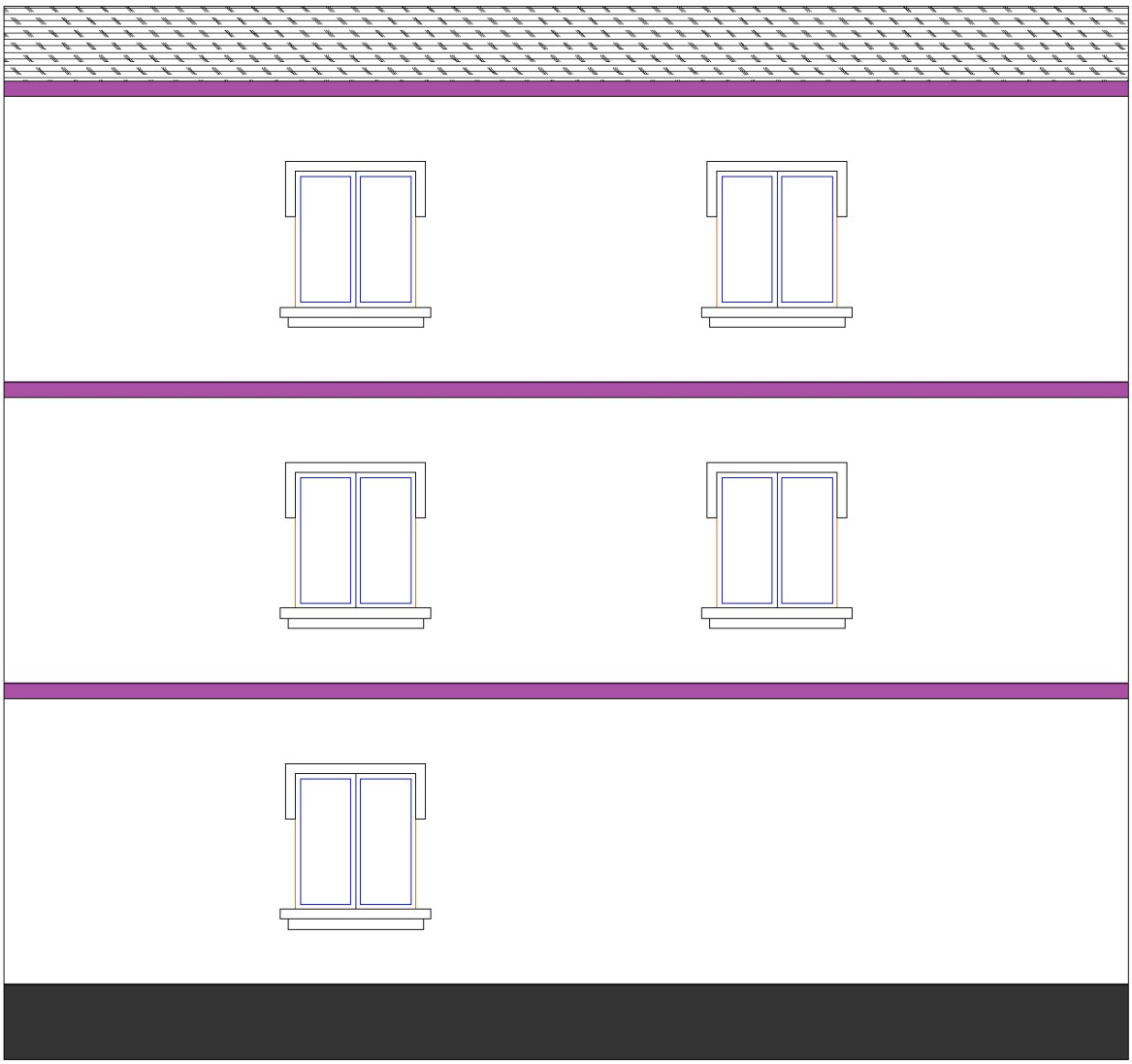
DESCRIPTION

Elevation of the front facade showing the three storeys, window openings, the entrance at ground level and the raised stair tower.

KEY INFORMATION FROM THE DRAWING

- Three storeys (ground + 2 upper floors)
- Stair tower rising above the roof line
- Window pattern repeated on each floor

Back Side Elevation



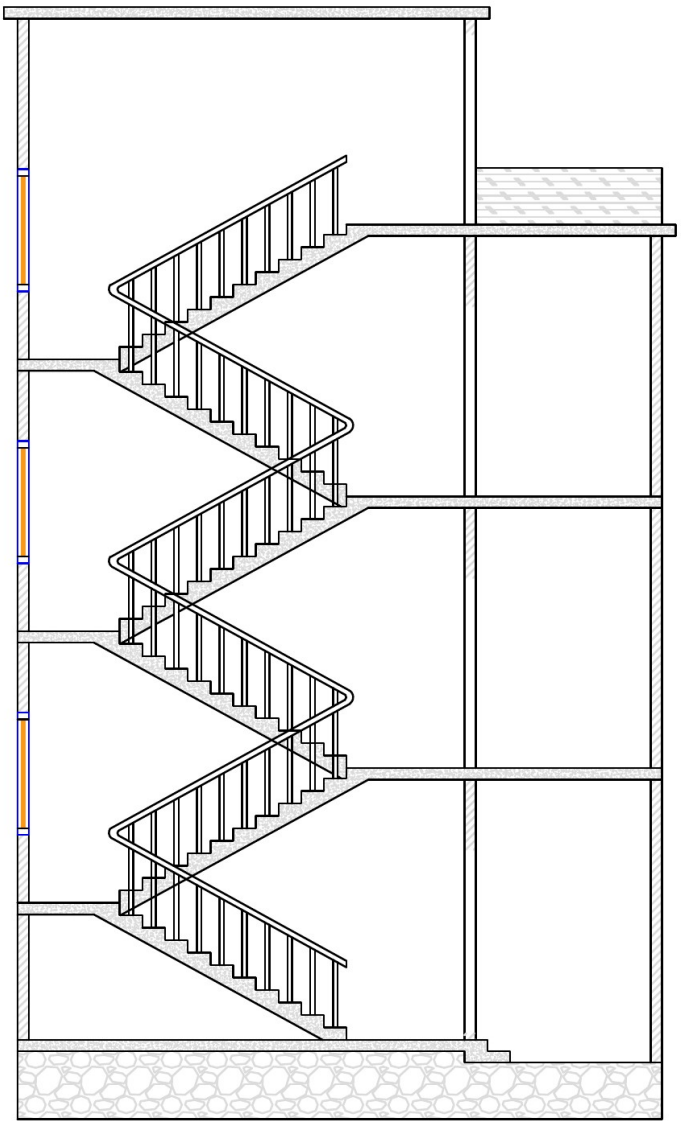
DESCRIPTION

Elevation of the rear facade showing the three storeys with the window openings on each floor.

KEY INFORMATION FROM THE DRAWING

- Three storeys, same height as the front elevation
- Two windows per floor on the upper levels

Cross Section (A-A)



CROSS SECTION (A-A)

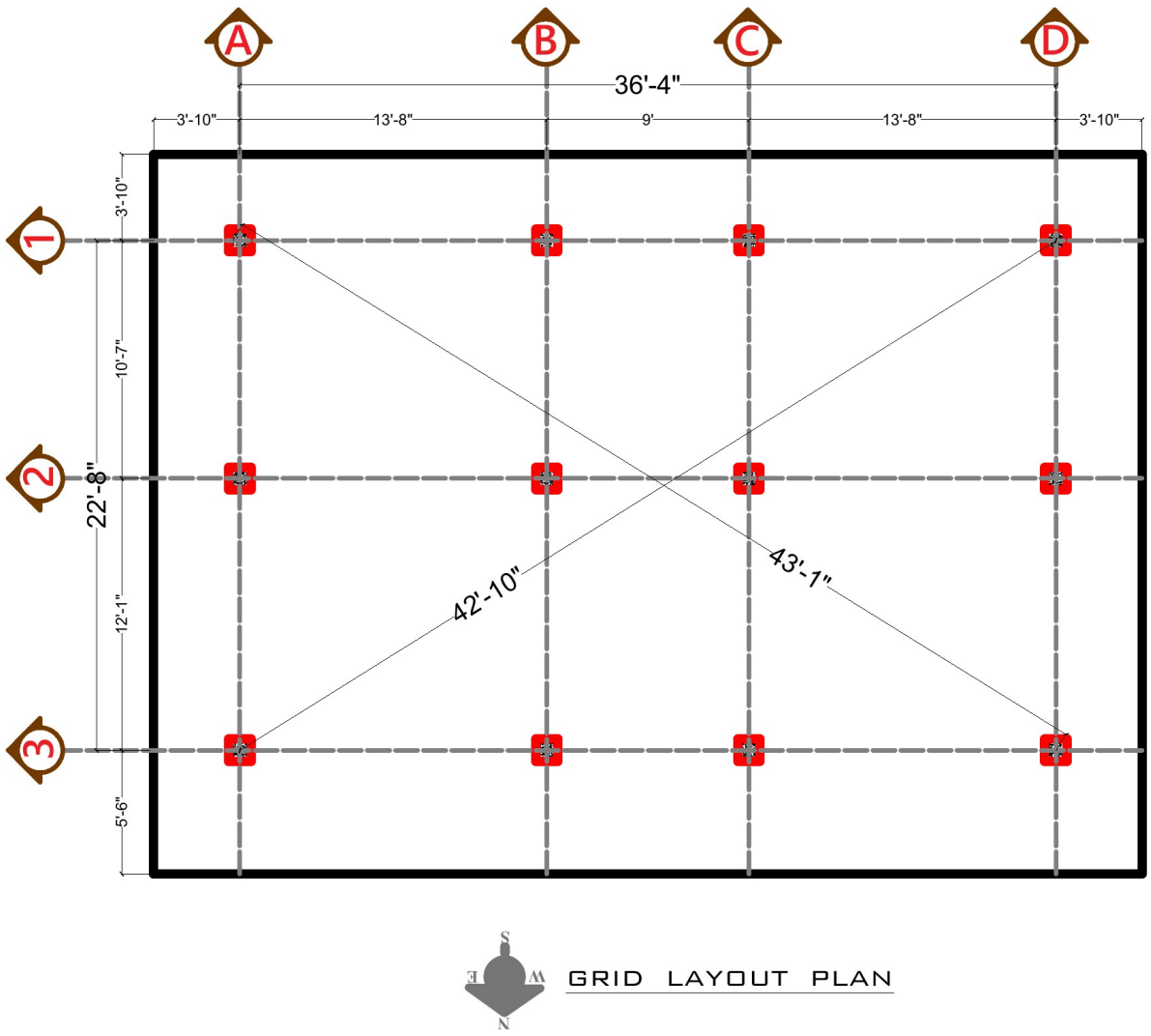
DESCRIPTION

Vertical section through the building along the stair, showing the floor levels, the stair flights and landings, and the roof slab.

KEY INFORMATION FROM THE DRAWING

- Stair flights and landings on all three levels
- Floor heights of 10' as dimensioned on the stair detail
- Roof slab at the top

Grid Layout Plan



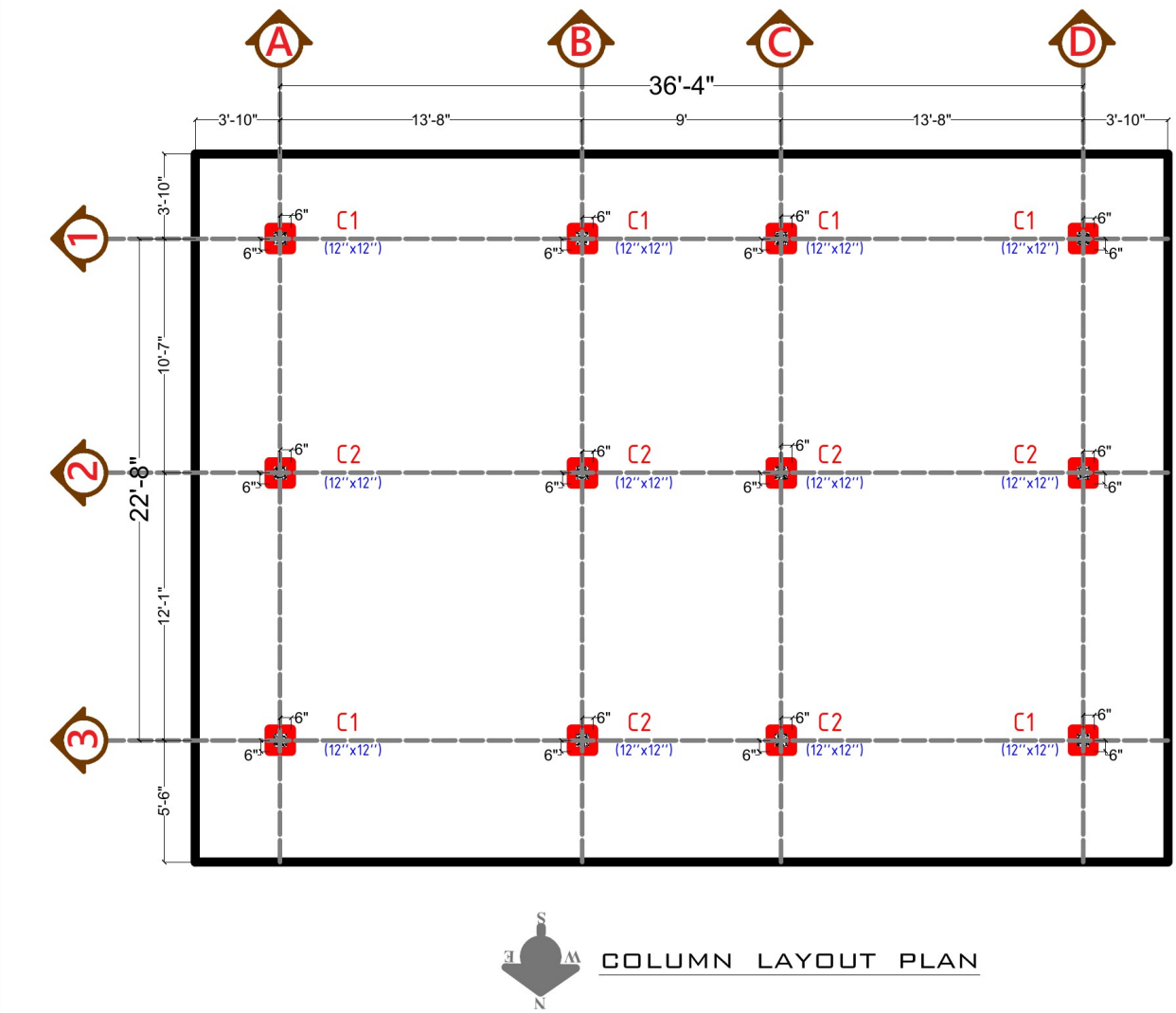
DESCRIPTION

Column grid used to set out the structure. Gridlines A to D run in one direction and 1 to 3 in the other, with spacing dimensions marked.

KEY INFORMATION FROM THE DRAWING

- Gridlines A-D and 1-3
- Overall size 36'-4" x 22'-8"
- Bay dimensions 13'-8", 9' and 13'-8" one way; 10'-7" and 12'-1" the other

Column Layout Plan



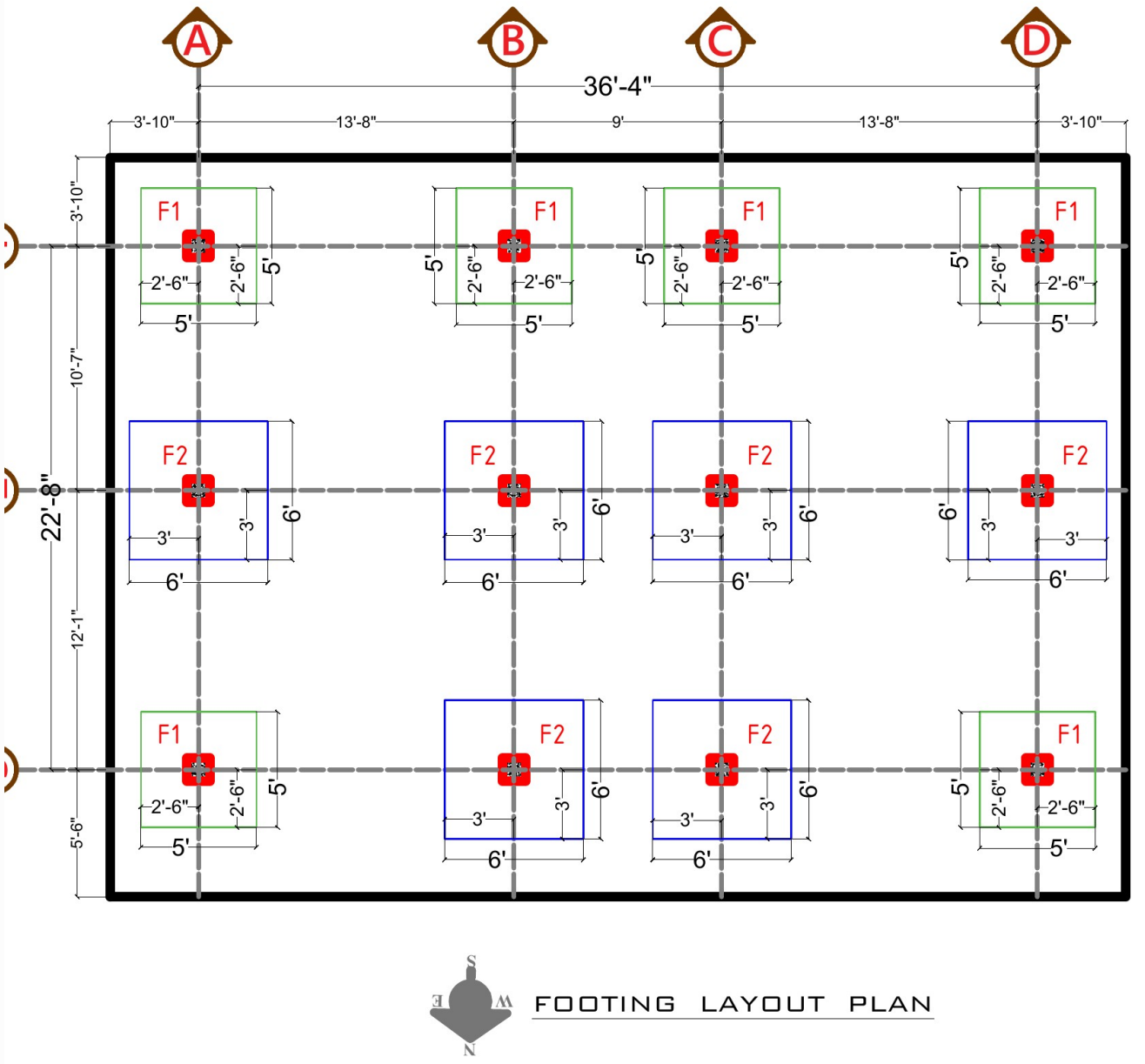
DESCRIPTION

Locations and marks of all columns on the grid. Two column types are used, each marked with its size.

KEY INFORMATION FROM THE DRAWING

- C1 and C2, both 12" x 12" at plan level
- Column bars and ties are on the Column Schedule sheet

Footing Layout Plan



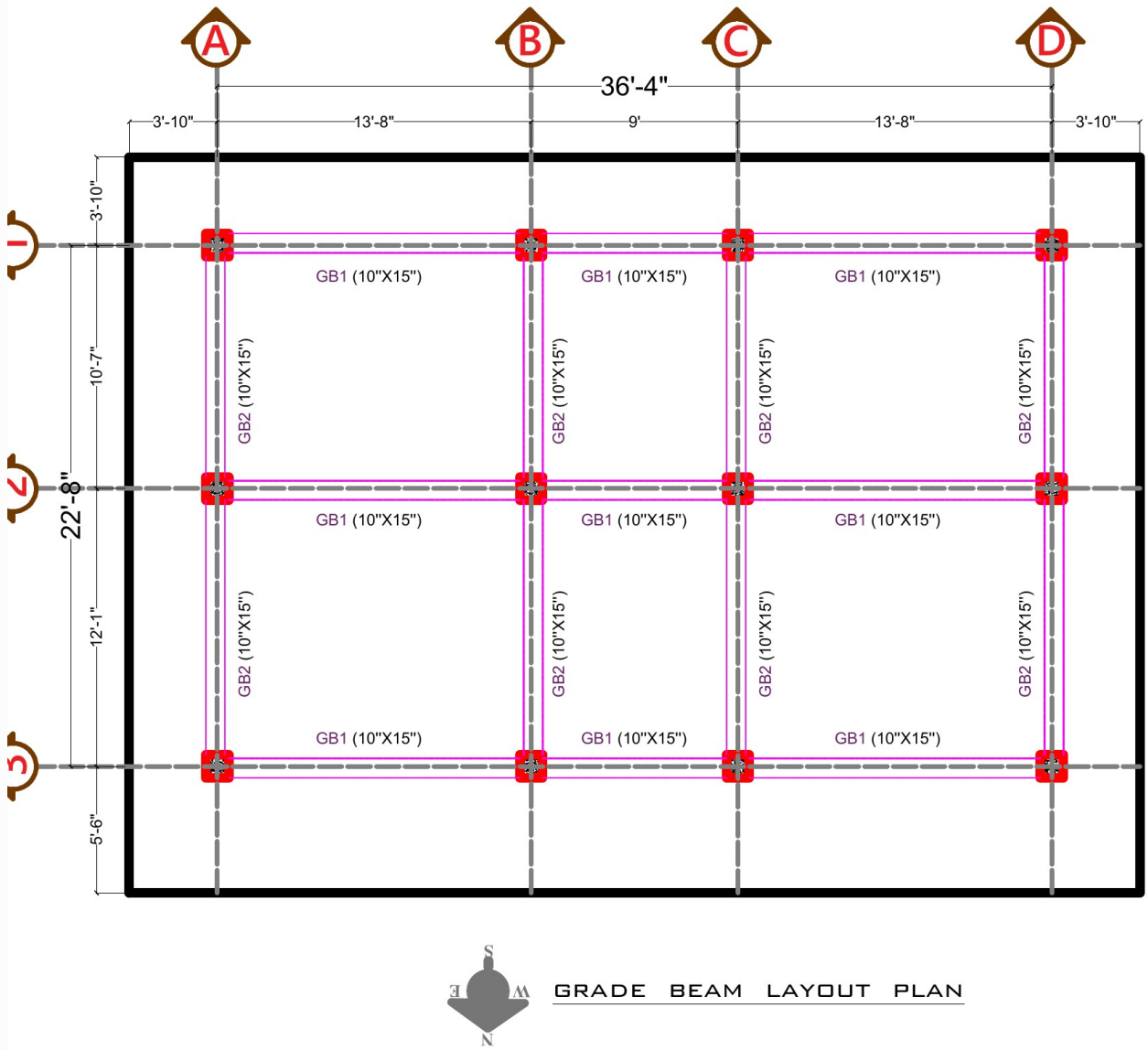
DESCRIPTION

Isolated footings below each column, marked F1 and F2 according to column load, with the footing sizes dimensioned.

KEY INFORMATION FROM THE DRAWING

- F1 5'-0" x 5'-0", 14" thick
- F2 6'-0" x 6'-0", 16" thick
- Same grid A-D and 1-3

Grade Beam Layout Plan



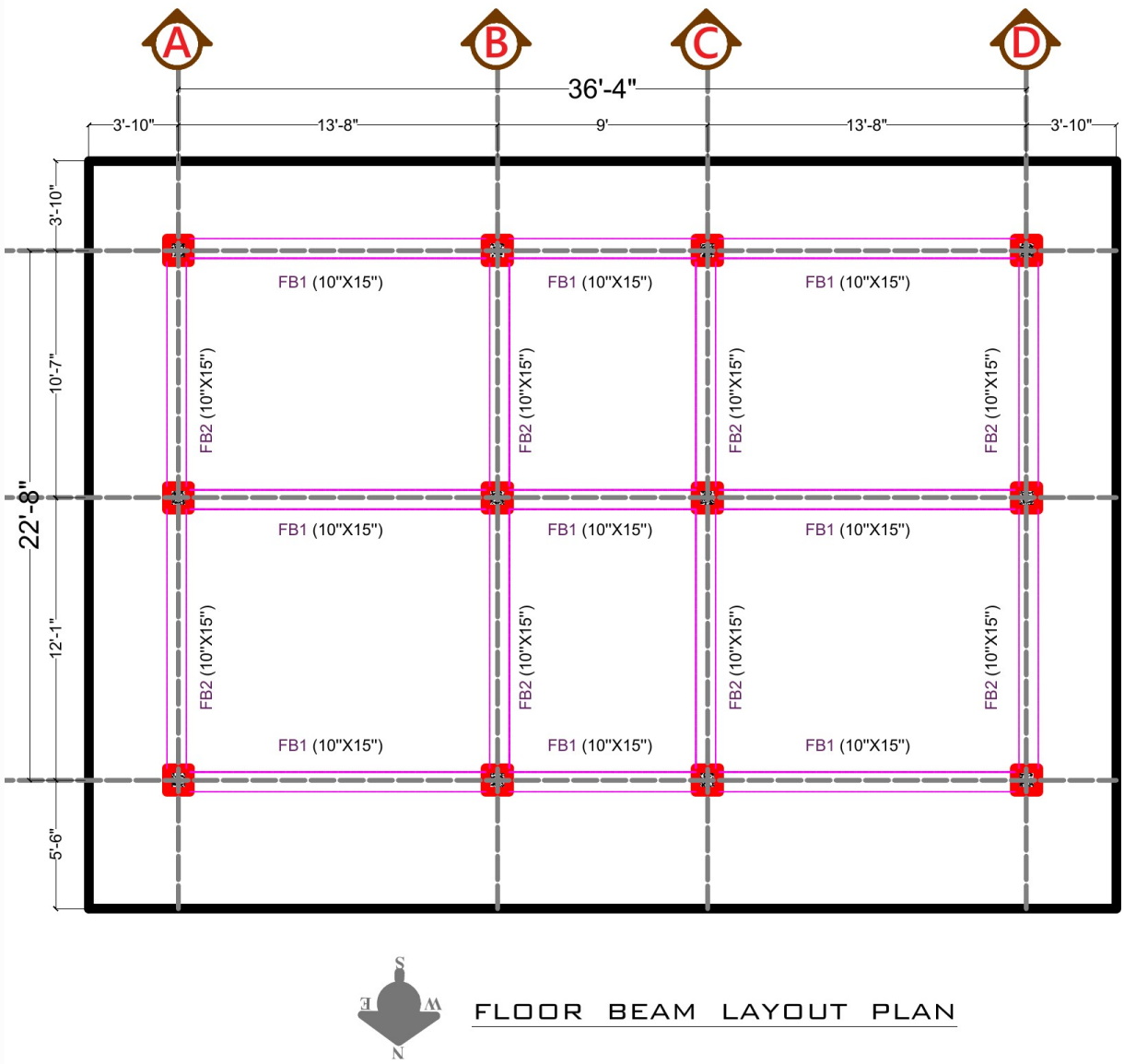
DESCRIPTION

Plan of the grade beams that tie the footings and columns together, with a mark and size on each beam.

KEY INFORMATION FROM THE DRAWING

- Beam marks GB1 and GB2
- All beams 10" x 15"
- Reinforcement is on the grade beam long sections

Floor Beam Layout Plan



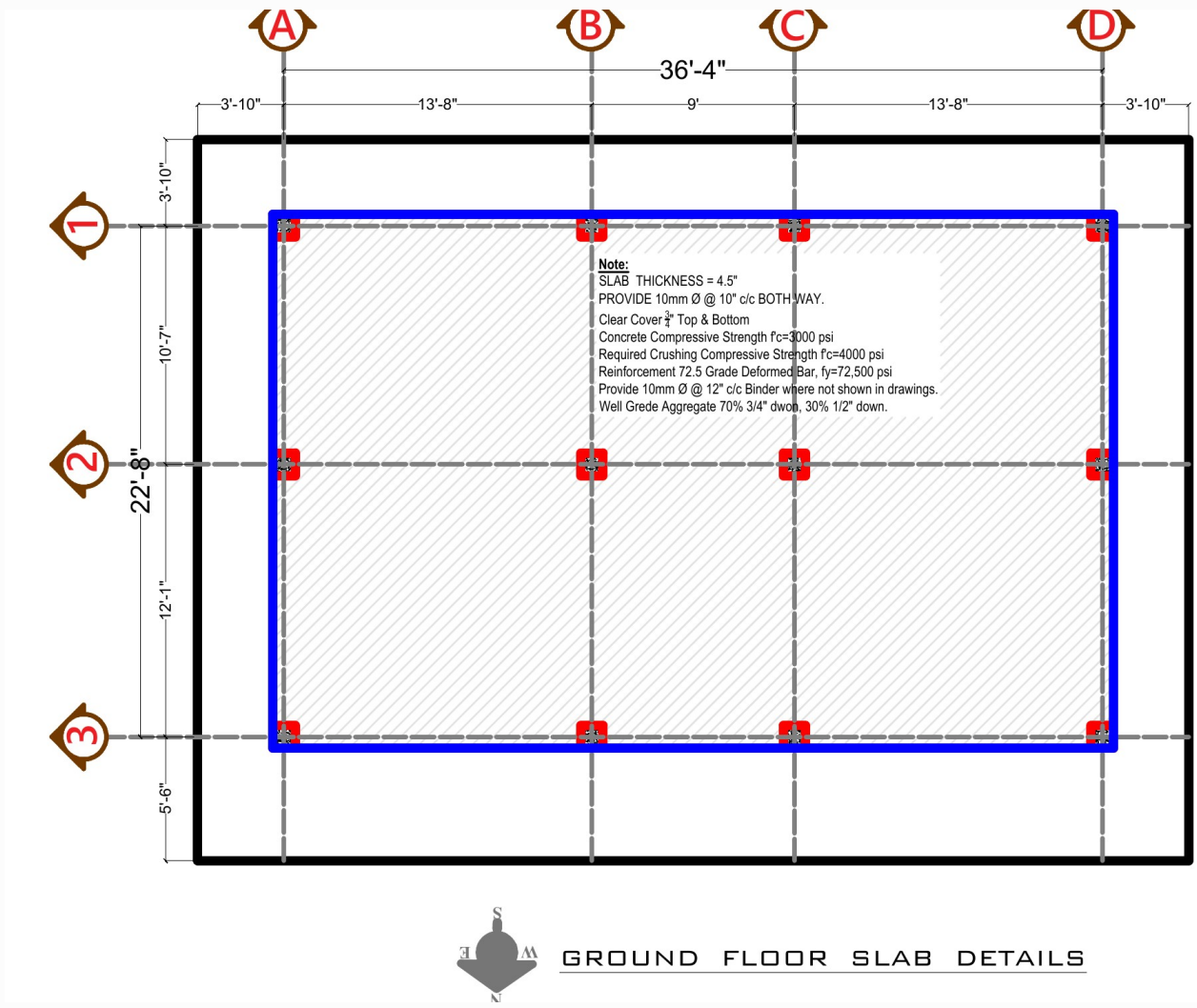
DESCRIPTION

Layout of the floor-level beams, each marked with its type and size.

KEY INFORMATION FROM THE DRAWING

- Beam marks FB1 and FB2
- All beams 10" x 15"
- Same grid and 36'-4" overall length

Ground Floor Slab Details



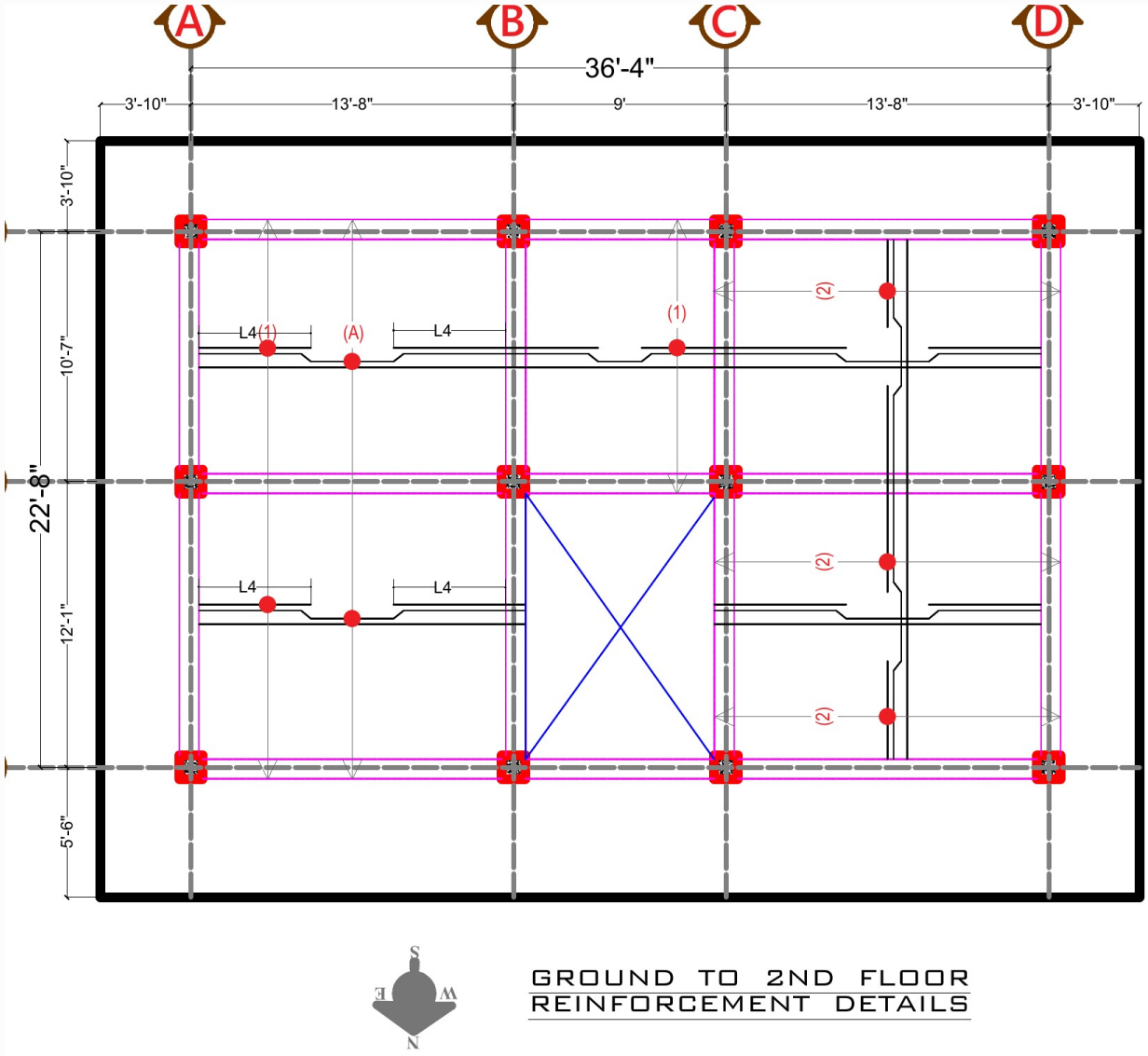
DESCRIPTION

Plan of the ground floor slab with the general notes for slab thickness, reinforcement and materials.

KEY INFORMATION FROM THE DRAWING

- Slab thickness 4.5"
- 10 mm dia bars @ 10" c/c both ways
- Clear cover 3/4" top and bottom
- Concrete f'c 3000 psi; required average strength 4000 psi
- Reinforcement 72.5 grade, fy 72,500 psi; binder 10 mm dia @ 12" c/c

Ground to 2nd Floor Reinforcement Details



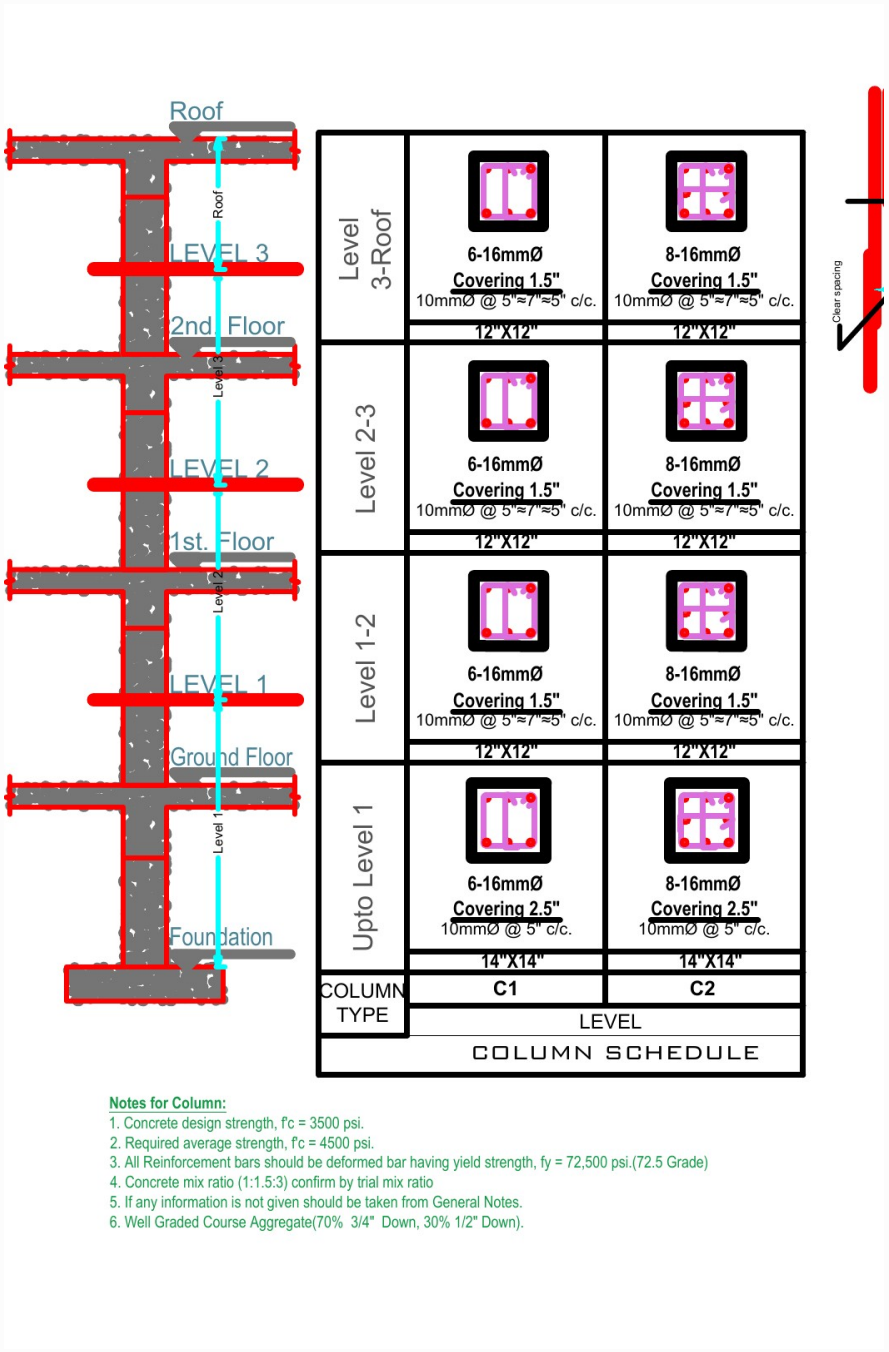
DESCRIPTION

Beam and slab reinforcement layout for the floors from ground to 2nd floor, with reinforcement zone marks and the stair opening.

KEY INFORMATION FROM THE DRAWING

- Stair opening shown with a crossed outline
- Reinforcement mark numbers (1), (2) and (A) on the panels
- L/4 zone marks along the beams

Column Schedule



Notes for Column:

1. Concrete design strength, f'c = 3500 psi.
2. Required average strength, f'c = 4500 psi.
3. All Reinforcement bars should be deformed bar having yield strength, fy = 72,500 psi.(72.5 Grade)
4. Concrete mix ratio (1:1.5:3) confirm by trial mix ratio
5. If any information is not given should be taken from General Notes.
6. Well Graded Course Aggregate(70% 3/4" Down, 30% 1/2" Down).

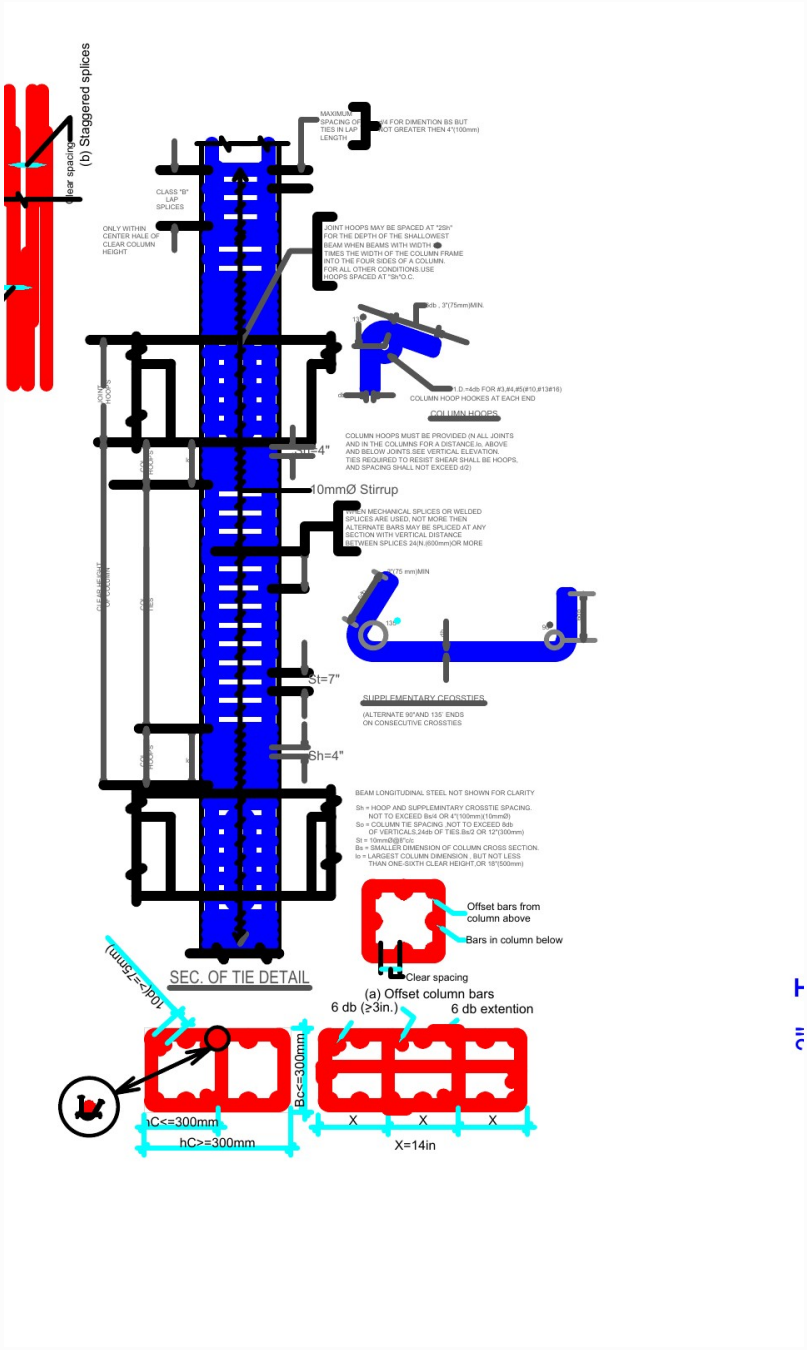
DESCRIPTION

Table of the column types by level, from the foundation to the roof. For each level it gives the section size, the longitudinal bars, the ties and the cover, with the notes for columns below.

KEY INFORMATION FROM THE DRAWING

- 12" x 12" sections, increasing to 14" x 14" at the lower level
- Longitudinal bars 6 nos or 8 nos of 16 mm dia
- Ties 10 mm dia @ 5" to 7" c/c
- Concrete f'c 3500 psi, required average 4500 psi, mix 1:1.5:3
- Reinforcement 72.5 grade, fy 72,500 psi

Column Tie and Splice Details



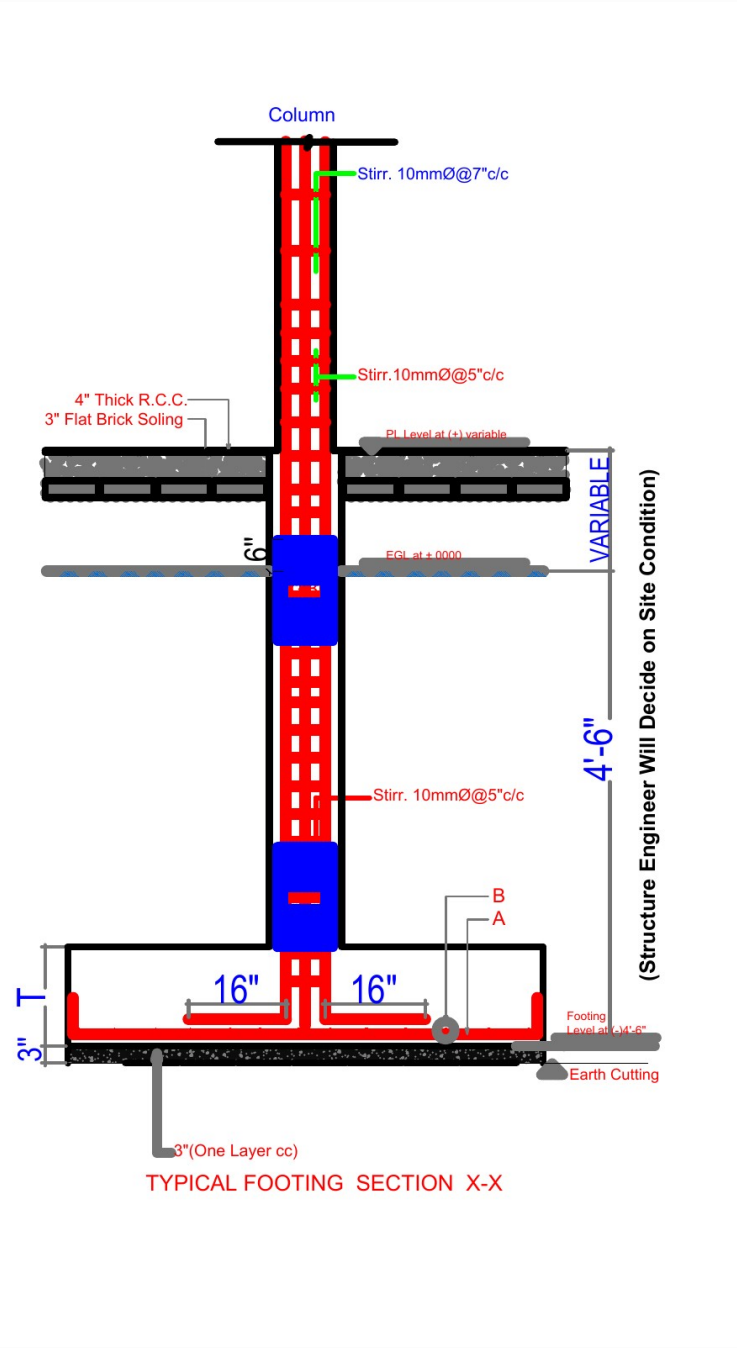
DESCRIPTION

Standard details for column ties and hoops, bar offsets and lap splices, with the spacing rules written beside each detail.

KEY INFORMATION FROM THE DRAWING

- Hoop spacing Sh not more than 4" near joints
- 135 degree hooks on column hoops
- Lap splices only within the middle half of the clear column height
- Offset and staggered bar splices shown

Typical Footing Section X-X



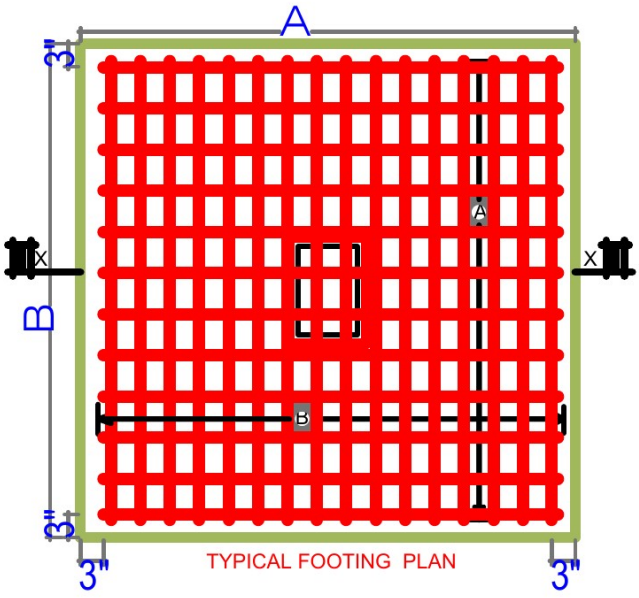
DESCRIPTION

Cross-section through a typical footing, showing the column starter bars and ties, the footing slab, the brick soling and the levels.

KEY INFORMATION FROM THE DRAWING

- Footing level (-)4'-6" below EGL (±0.000)
- 3" flat brick soling and 4" thick R.C.C. layer
- Column ties 10 mm dia @ 5" and @ 7" c/c
- Footing level to be decided on site by the structural engineer

Footing Schedule and Typical Footing Plan



FOOTING SCHEDULE :

Sl. No.	L (Ft.)	B (Ft.)	T	A Mainbar	B Binder
F_1	5'-0"	5'-0"	14"	16 mmØ@ 6"c/c	16 mmØ@ 6"c/c
F_2	6'-0"	6'-0"	16"	16 mmØ@ 6"c/c	16 mmØ@ 6"c/c

Note :
1). Clear Cover of the Footing will be 3" in all sides
2). Concrete design strength, f'c = 3000 psi.
3). Required average strength, f'c = 4000 psi.
4). All Reinforcement bars should be deformed bar having yield strength, fy = 72,500 psi.(72.5 Grade)
5). Concrete mix ratio (1:1.5:3) confirm by trial mix ratio
6). If any information is not given should be taken from General Notes.
8). Well Graded Course Aggregate(70% 3/4" Down, 30% 1/2" Down).
9). A = Perpendicular Bar To Long Side Of Column (Lower Layer)
10). B = Perpendicular Bar To Short Side Of Column (Upper Layer)

FOOTING SCHEDULE

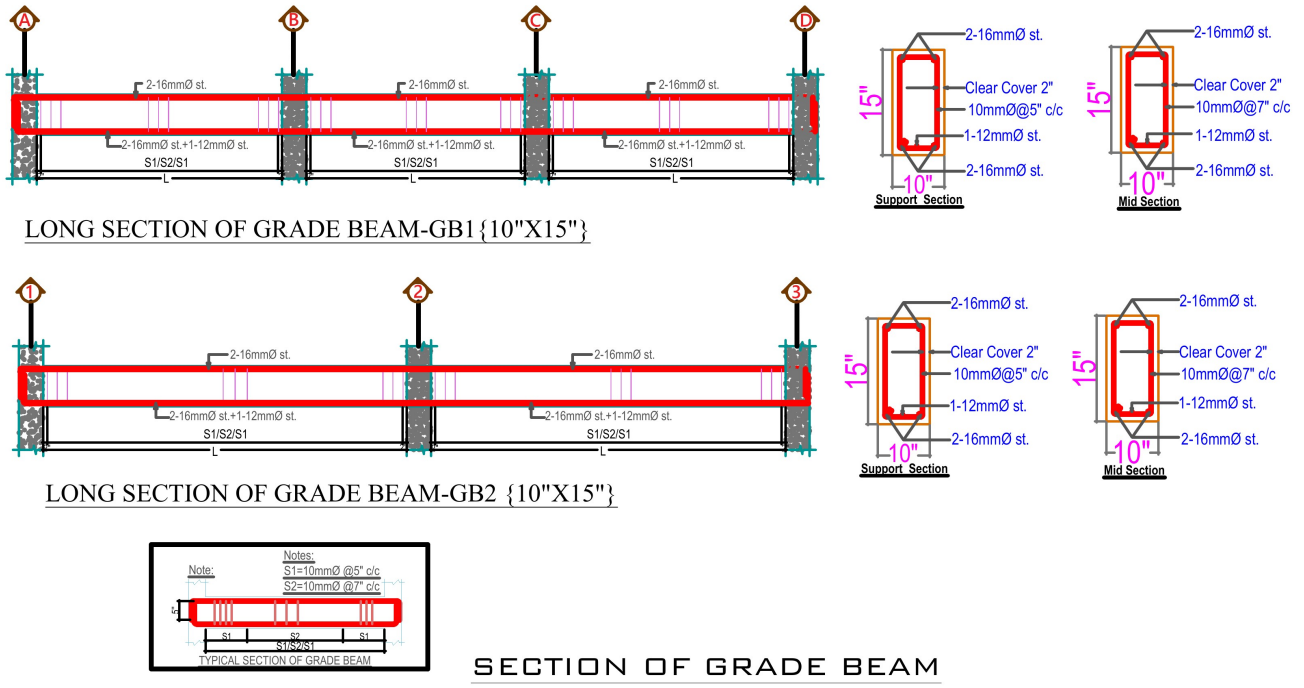
DESCRIPTION

Typical footing plan with the bar mesh, together with the footing schedule table giving the size, depth and bars of each footing and the notes that apply to all.

KEY INFORMATION FROM THE DRAWING

- F1 5' x 5' x 14" and F2 6' x 6' x 16"
- Bars 16 mm dia @ 6" c/c in both layers
- Clear cover 3" on all sides
- Concrete f'c 3000 psi, required average 4000 psi; fy 72,500 psi

Grade Beam Long Sections GB1 and GB2, Typical Section



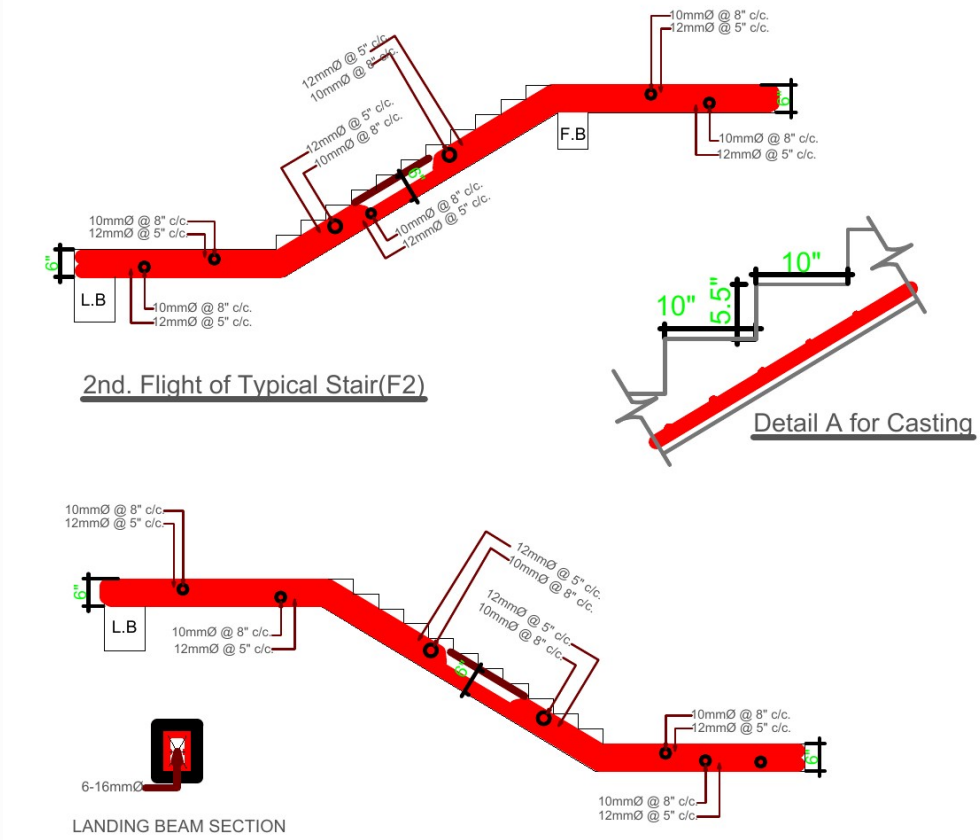
DESCRIPTION

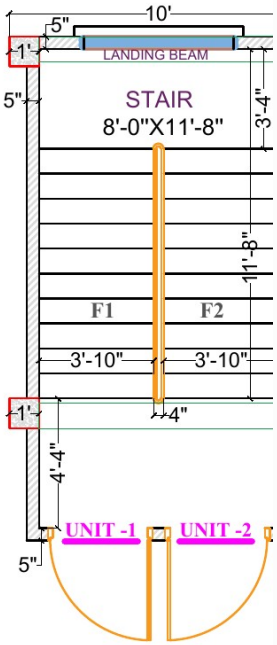
Longitudinal sections of grade beams GB1 and GB2 with the support and mid sections, showing the main bars and stirrup spacing, plus the typical grade beam section.

KEY INFORMATION FROM THE DRAWING

- Beam size 10" x 15", clear cover 2"
- 2-16 mm dia straight bars, with an added 1-12 mm dia bar in GB2
- Stirrups 10 mm dia @ 5" c/c at supports and @ 7" c/c in the middle

Stair Details





Stair Details

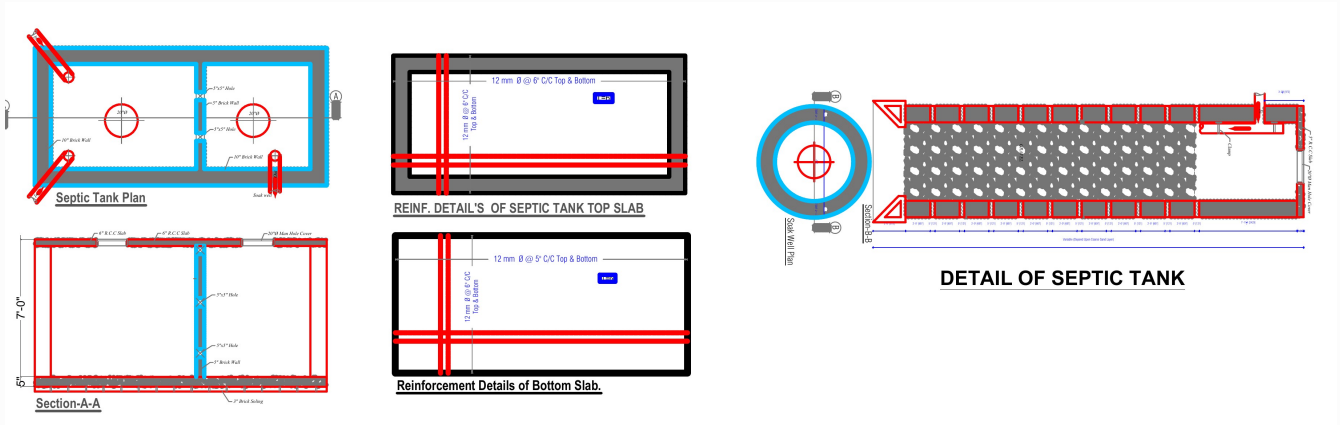
DESCRIPTION

Typical stair details: the first and second flights with their reinforcement, the casting detail, the landing beam section and the stair plan.

KEY INFORMATION FROM THE DRAWING

- Flights F1 and F2; stair 8'-0" x 11'-8"
- Main bars 12 mm dia @ 5" c/c, distribution 10 mm dia @ 8" c/c
- Landing beam with 6-16 mm dia bars

Septic Tank and Soak Well Details



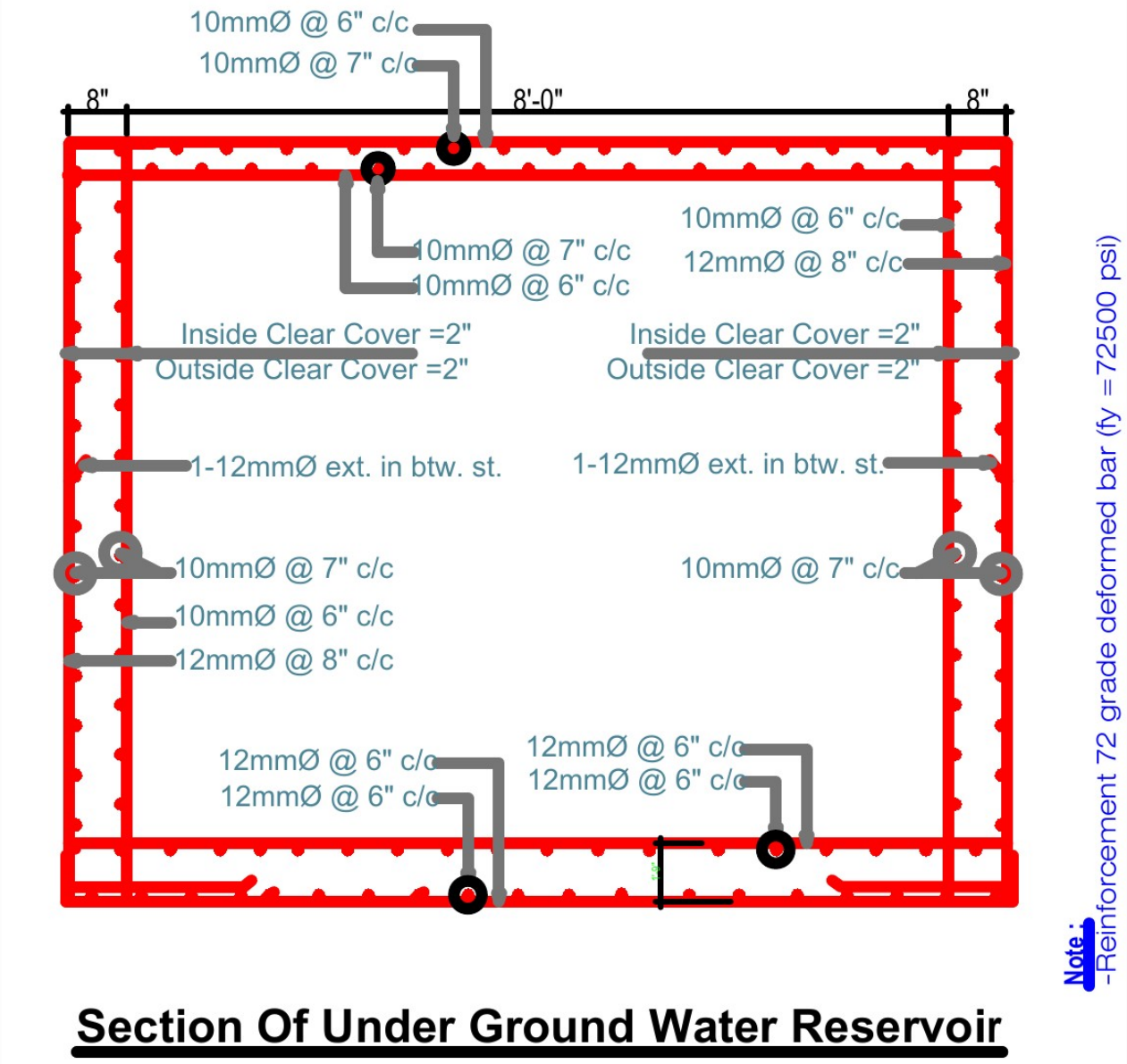
DESCRIPTION

Plans, sections and reinforcement details for the septic tank and the soak well.

KEY INFORMATION FROM THE DRAWING

- Top and bottom slabs 6" thick with 12 mm dia bars
- Brick walls 5" and 10" thick
- Soak well with gravel fill and 5" brick rings
- Reinforcement 72 grade; concrete f'c 3,500 psi

Section of Underground Water Reservoir



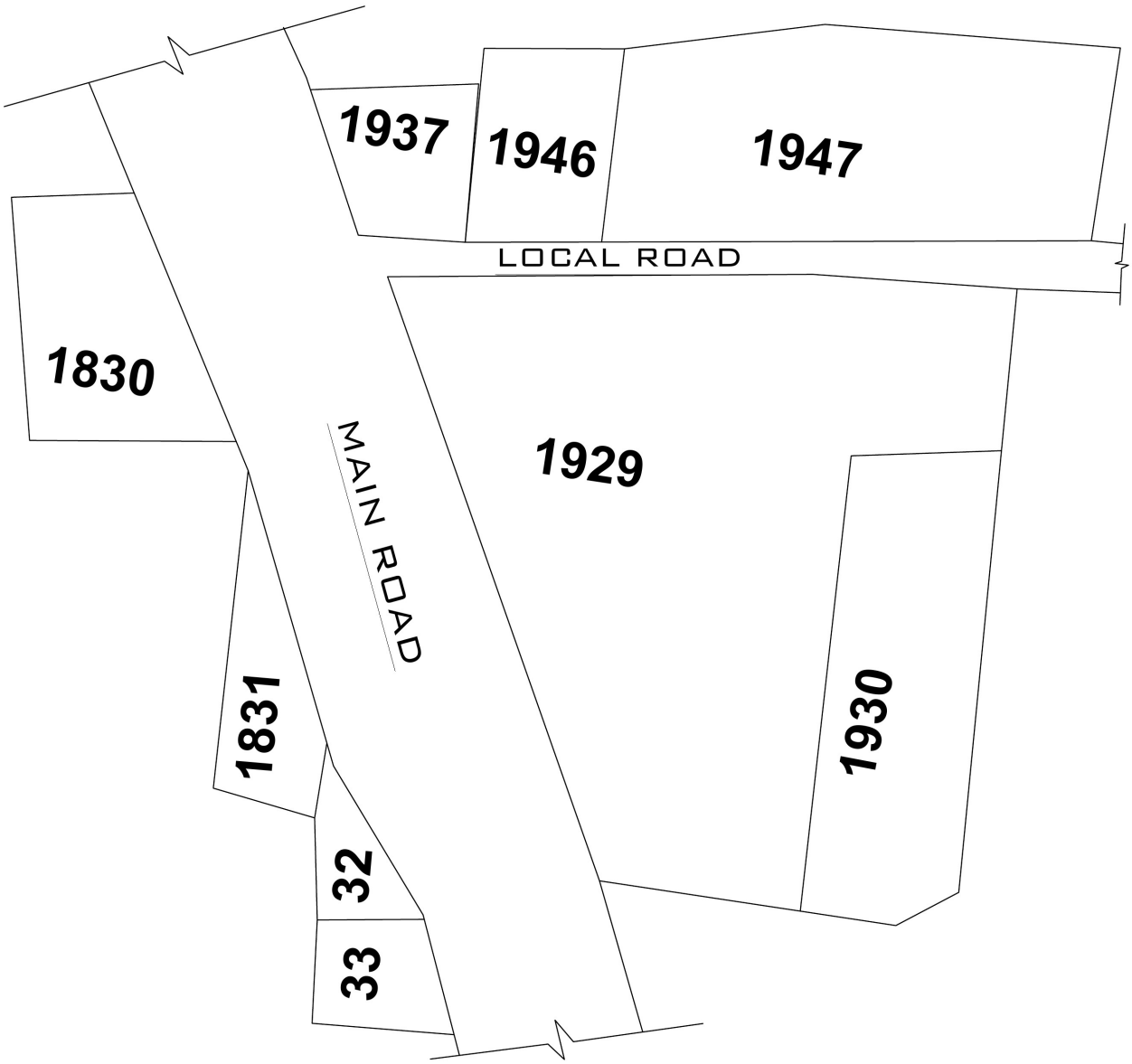
DESCRIPTION

Section through the underground water reservoir showing the wall and slab reinforcement and the clear covers.

KEY INFORMATION FROM THE DRAWING

- Wall bars 10 mm dia @ 6" and @ 7" c/c; slab bars 12 mm dia @ 6" and @ 8" c/c
- Clear cover 2" inside and outside
- Concrete f'c 3,500 psi; 72 grade bars, fy 72,500 psi
- Damp-proof admixture in the concrete

Plot Map



DESCRIPTION

Map of the surrounding plots with their dag numbers, the main road and a local road. The sheet gives this map no title, and the project plot is not marked on it.

KEY INFORMATION FROM THE DRAWING

- Plots 1829, 1830, 1831, 1929, 1930, 1937, 1946 and 1947, and 32 and 33
- Main Road and Local Road shown
- Not drawn to scale